A negative leading coefficient in the third LIS iterate

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Abstract

The longest-increasing-subsequence polynomials are not infinitely q-log-convex. Their third log-convexity iterate at index 5 has degree 34 and leading coefficient -32. Only the top three coefficients of $P_2, \dots, P_8$ are needed, giving a short exact disproof of Chen's Conjecture 1.4.

Task 1. Full disproof

Let $P_{n,k}$ count permutations of $\{1, \dots, n\}$ whose longest increasing subsequence has length k, and put

$$P_n(x) = \sum_{k=1}^{n} P_{n,k} x^k.$$

Following [1, p. 5], use the centered operator

$$(\mathcal{H}A)_n = A_{n-1} A_{n+1} - A_n^2.$$

Infinite q-log-convexity means coefficientwise nonnegativity of every iterate. The letter x plays the role of q.

Theorem 1. *The third iterate satisfies*

$$(\mathcal{H}^3 P)_5(x) = -32x^{34} + \textit{a polynomial of degree at most } 33. \tag{1}$$

Consequently Chen's infinite q-log-convexity conjecture is false.

Proof. Write the top three coefficients as

$$P_n(x) = x^n + a_n x^{n-1} + b_n x^{n-2} + \text{lower-degree terms}.$$

The needed values, also printed in [1, pp. 2–3], are

n	2	3	4	5	6	7	8
a_n	1	4	9	16	25	36	49
b_n	0	1	13	61	181	421	841

(2)

These counts can be obtained without enumerating all permutations. The Robinson–Schensted correspondence gives $P_{n,k} = \sum_{\lambda \vdash n,\, \lambda_1 = k} (f^\lambda)^2$, where f^λ counts standard Young tableaux of shape

λ. For $k = n-1$ there is just the shape $(n-1,1)$, so $a_n = (n-1)^2$. For $n \geq 4$ and $k = n-2$ the shapes are $(n-2,2)$ and $(n-2,1,1)$. The hook-length formula gives

$$b_n = \left(\frac{n(n-3)}{2}\right)^2 + \left(\frac{(n-1)(n-2)}{2}\right)^2.$$

The two smaller values are $b_2 = 0$ and $b_3 = 1$. This proves (2).

Put $B_n = (\mathcal{H}P)_n$ and $C_n = (\mathcal{H}^2P)_n$. The leading terms of degree $2n$ cancel in B_n. The next coefficient is $a_{n-1} + a_{n+1} - 2a_n = 2$. Thus, for $3 \leq n \leq 7$,

$$B_n(x) = 2x^{2n-1} + v_n x^{2n-2} + \text{lower-degree terms},$$

where multiplication gives

$$v_n = b_{n-1} + b_{n+1} + a_{n-1}a_{n+1} - a_n^2 - 2b_n.$$

The complete required table is

n	3	4	5	6	7
v_n	4	19	41	71	109

(3)

In $C_n = B_{n-1}B_{n+1} - B_n^2$, the terms of degree $4n-2$ again cancel. The coefficient of degree $4n-3$ is

$$c_n = 2(v_{n-1} + v_{n+1} - 2v_n).$$

Using (3),

$$C_4 = 14x^{13} + \cdots, \qquad C_5 = 16x^{17} + \cdots, \qquad C_6 = 16x^{21} + \cdots. \tag{4}$$

Therefore $C_4C_6 - C_5^2$ has degree 34 and leading coefficient $14 \cdot 16 - 16^2 = -32$, proving (1). □

The witness uses only $P_2, \ldots, P_8$; it involves no boundary convention for P_0. It also gives a failure at an ordinary positive parameter. Direct evaluation of the same polynomials yields

$$(\mathcal{H}^3P)_5(6) = -844900534849808869264908288000 0 < 0. \tag{5}$$

The exact numeric iterate table is included in the verification output.

Verification and scope

The verifier enumerates all 46232 permutations of sizes 2 through 8 and checks each longest-increasing-subsequence length by two algorithms. It independently reconstructs all seven polynomials by the hook-length formula, checks the full polynomial iterates by two exact multiplication methods, and verifies (5). No floating point is used. Review is by the authoring agent only.

The source's reported computational evidence for third-order q-log-convexity on p. 5 is incompatible with (1). The definition and printed input polynomials themselves agree with the independent enumeration. This is a counterexample to the intended conjecture, not a result based on a missing endpoint or a transcription error. It does not decide ordinary q-log-convexity, second-order q-log-convexity, or infinite log-concavity of the coefficient sequence of each individual P_n.

Tasks 2–5. Follow-up directions

2. Decide whether $(P_n(x))_{n\geq 1}$ is second-order q-log-convex.

3. Determine the positive parameters x for which the numerical sequence $(P_n(x))_{n\geq 1}$ is infinitely log-convex.

4. Locate all negative coefficients in higher iterates and determine whether each fixed iterate is eventually coefficientwise nonnegative as n increases.

5. Derive formulas for the leading nonzero coefficients of $\mathcal{H}^r P$ for general r, using tableaux close to a single row.

References

[1] W. Y. C. Chen, *Log-concavity and q-Log-convexity Conjectures on the Longest Increasing Subsequences of Permutations*, arXiv:0806.3392v2, June 22, 2008. Conjecture 1.4, pp. 1–5.

[2] M. Bóna, M.-L. Lackner and B. E. Sagan, *Longest Increasing Subsequences and Log Concavity*, Ann. Comb. 21 (2017), 535–549, doi:10.1007/s00026-017-0365-x. Sections 6.3–6.4 distinguish related conjectures.

[3] MathDB, Problem 317479, accessed September 18, 2026.